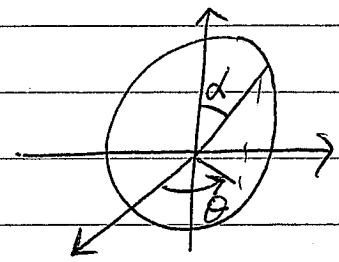
$\boxed{P2}$ let $\vec{S} = (\cos\theta \sin\alpha, \sin\theta \sin\alpha, \cos\alpha) =$

$$= \left(\sqrt{1-\sigma^2}\cos\theta, \sqrt{1-\sigma^2}\sin\theta, \sigma\right)$$

where $\sigma = \cos\alpha$



Then $Q = \frac{1}{8\pi}\int d^2x\, \varepsilon_{\mu\nu}\, \vec{S}\cdot \partial_\mu \vec{S} \times \partial_\nu \vec{S} =$

$$= \frac{1}{8\pi}\int d^2x\, \varepsilon_{\mu\nu}\, \varepsilon_{abc}\, S_a\, \partial_\mu S_b\, \partial_\nu S_c \qquad (33)$$

$$Q = \frac{1}{8\pi}\int d^2x \; \varepsilon_{abc} \; \varepsilon_{\mu\nu} \; S_a \; \partial_\mu S_b \; \partial_\nu S_c =$$

$$= \frac{1}{8\pi}\int d^2x \; \varepsilon_{123} \; \varepsilon_{\mu\nu} \; S_1\left(\partial_\mu S_2 \, \partial_\nu S_3 - \partial_\mu S_3 \, \partial_\nu S_2\right)$$
$$+ \varepsilon_{231} \; \varepsilon_{\mu\nu} \; S_2 \left(\partial_\mu S_3 \, \partial_\nu S_1 - \partial_\mu S_1 \, \partial_\nu S_3\right)$$
$$+ \varepsilon_{312} \; \varepsilon_{\mu\nu} \; S_3 \left(\partial_\mu S_1 \, \partial_\nu S_2 - \partial_\mu S_2 \, \partial_\nu S_1\right)$$

$$S_1 = \sqrt{1-\sigma^2}\,\cos\theta \;, \quad S_2 = \sqrt{1-\sigma^2}\,\sin\theta \;, \quad S_3 = \sigma$$

$$\partial_\mu S_1 = -\frac{\sigma\cos\theta}{\sqrt{1-\sigma^2}}\,\partial_\mu\sigma - \sqrt{1-\sigma^2}\,\sin\theta\,\partial_\mu\theta$$

$$\partial_\mu S_2 = -\frac{\sigma\sin\theta}{\sqrt{1-\sigma^2}}\,\partial_\mu\sigma + \sqrt{1-\sigma^2}\,\cos\theta\,\partial_\mu\theta \;; \quad \partial_\mu S_3 = \partial_\mu\sigma$$

$$\Rightarrow S_1\left(\partial_\mu S_2\,\partial_\nu S_3 - \partial_\mu S_3\,\partial_\nu S_2\right) = \sqrt{1-\sigma^2}\,\cos\theta\left\{\left(-\frac{\cos\theta\,\sigma}{\sqrt{1-\sigma^2}}\,\partial_\mu\sigma + \sqrt{1-\sigma^2}\,\sin\theta\,\partial_\mu\theta\right)\partial_\nu\sigma\right.$$
$$\left. - \partial_\mu\sigma\left(-\frac{\cos\theta\,\sigma}{\sqrt{1-\sigma^2}}\,\partial_\nu\sigma + \sqrt{1-\sigma^2}\,\sin\theta\,\partial_\nu\theta\right)\right\} = \left(1-\sigma^2\right)\cos^2\theta\left(\partial_\mu\theta\,\partial_\nu\sigma - \partial_\mu\sigma\,\partial_\nu\theta\right)$$

$$\cancel{\varepsilon_{\mu\nu}\,\partial_\mu S_2\,\partial_\nu S_3 = \partial_\mu S_3\,\partial_\nu S_2}$$

$$S_2\left(\partial_\mu S_3\,\partial_\nu S_1 - \partial_\mu S_1\,\partial_\nu S_3\right) = \sqrt{1-\sigma^2}\,\sin\theta\left\{\partial_\mu\sigma\left(-\frac{\cos\theta\,\sigma}{\sqrt{1-\sigma^2}}\,\partial_\nu\sigma - \sqrt{1-\sigma^2}\,\sin\theta\,\partial_\nu\theta\right)\right.$$
$$\left. - \left(-\frac{\cos\theta\,\sigma}{\sqrt{1-\sigma^2}}\,\partial_\mu\sigma - \sqrt{1-\sigma^2}\,\sin\theta\,\partial_\mu\theta\right)\partial_\nu\sigma\right\} = \left(1-\sigma^2\right)\sin^2\theta\left(\partial_\mu\theta\,\partial_\nu\sigma - \partial_\mu\sigma\,\partial_\nu\theta\right)$$

$$S_3 = \sigma\left\{\left(-\frac{\sigma\cos\theta}{\sqrt{1-\sigma^2}}\,\partial_\mu\sigma - \sqrt{1-\sigma^2}\,\sin\theta\,\partial_\mu\theta\right)\left(-\frac{\sigma\sin\theta}{\sqrt{1-\sigma^2}}\,\partial_\nu\sigma + \sqrt{1-\sigma^2}\,\cos\theta\,\partial_\nu\theta\right)\right.$$
$$\left. - \left(-\frac{\sigma\sin\theta}{\sqrt{1-\sigma^2}}\,\partial_\mu\sigma + \sqrt{1-\sigma^2}\,\cos\theta\,\partial_\mu\theta\right)\left(-\frac{\sigma\cos\theta}{\sqrt{1-\sigma^2}}\,\partial_\nu\sigma - \sqrt{1-\sigma^2}\,\sin\theta\,\partial_\nu\theta\right)\right\}$$

$$= \sigma\left\{-\sigma\cos^2\theta\,\partial_\mu\sigma\,\partial_\nu\theta + \sigma\sin^2\theta\,\partial_\mu\theta\,\partial_\nu\sigma - \sigma\sin^2\theta\,\partial_\mu\sigma\,\partial_\nu\theta + \sigma\cos^2\theta\,\partial_\mu\theta\,\partial_\nu\sigma\right\}$$

$$= \sigma^2\left(\partial_\mu\theta\,\partial_\nu\sigma - \partial_\mu\sigma\,\partial_\nu\theta\right)\left(\sin^2\theta + \cos^2\theta\right) = \sigma^2\left(\partial_\mu\theta\,\partial_\nu\sigma - \partial_\mu\sigma\,\partial_\nu\theta\right)$$

$$\Rightarrow Q = \frac{1}{8\pi}\int d^2x \; \varepsilon_{\mu\nu}\left[\left(1-\sigma^2\right)\cos^2\theta + \left(1-\sigma^2\right)\sin^2\theta + \sigma^2\right]\left(\partial_\mu\theta\,\partial_\nu\sigma - \partial_\mu\sigma\,\partial_\nu\theta\right)$$

$$= \frac{1}{8\pi}\int d^2x\left(\varepsilon_{\mu\nu}\,\partial_\mu\theta\,\partial_\nu\sigma - \underbrace{\varepsilon_{\mu\nu}\,\partial_\mu\sigma\,\partial_\nu\theta}_{-\varepsilon_{\mu\nu}\,\partial_\nu\theta\,\partial_\mu\sigma}\right) =$$

$$= \frac{1}{4\pi}\int d^2x \; \varepsilon_{\mu\nu}\,\partial_\mu\theta\,\partial_\nu\sigma \,.$$

$$Q = \frac{1}{4\pi}\int d^2x\ \varepsilon_{\mu\nu}\ \partial_\mu\theta\ \partial_\nu\sigma = \frac{1}{4\pi}\int d^2x\ \varepsilon_{\mu\nu}\ \partial_\mu\theta\left(-\sin\alpha\ \partial_\nu\alpha\right)$$

$$= \frac{1}{4\pi}\int d^2x\ \varepsilon_{\nu\mu}\ \sin\alpha\ \partial_\nu\alpha\ \partial_\mu\theta \qquad (34)$$

Since $\theta = \theta(\phi)$ while $\alpha = \alpha(r)$

$$Q = \frac{1}{4\pi}\int d^2x\ \sin\alpha\left(\vec{\nabla}\alpha \times \vec{\nabla}\theta\right)_z$$

$$\begin{cases} \vec{\nabla}\alpha = \dfrac{d\alpha}{dr}\ \hat{r} \\ \vec{\nabla}\theta = \dfrac{1}{r}\dfrac{d\theta}{d\phi}\ \hat{\phi} = \dfrac{q}{r}\ \hat{\phi} \end{cases} \qquad \left(\vec{r}\times\hat{\phi} = \hat{z}\right)$$

$$Q = \frac{1}{4\pi}\int_0^\infty dr\cdot r\int_0^{2\pi} d\phi\ \sin\alpha(r)\ \frac{d\alpha}{dr}\ \frac{q}{r} = \frac{q}{4\pi}\int_0^\infty d(-\cos\alpha)\int_0^{2\pi} d\phi$$

$$= -\frac{q}{2}\Big(\underbrace{\cos\alpha(r=\infty)}_{\frac{\pi}{2}} - \underbrace{\cos\alpha(r=0)}_{0\ \text{or}\ \pi}\Big) = \frac{q}{2}\left(\pm 1\right). \qquad (35)$$